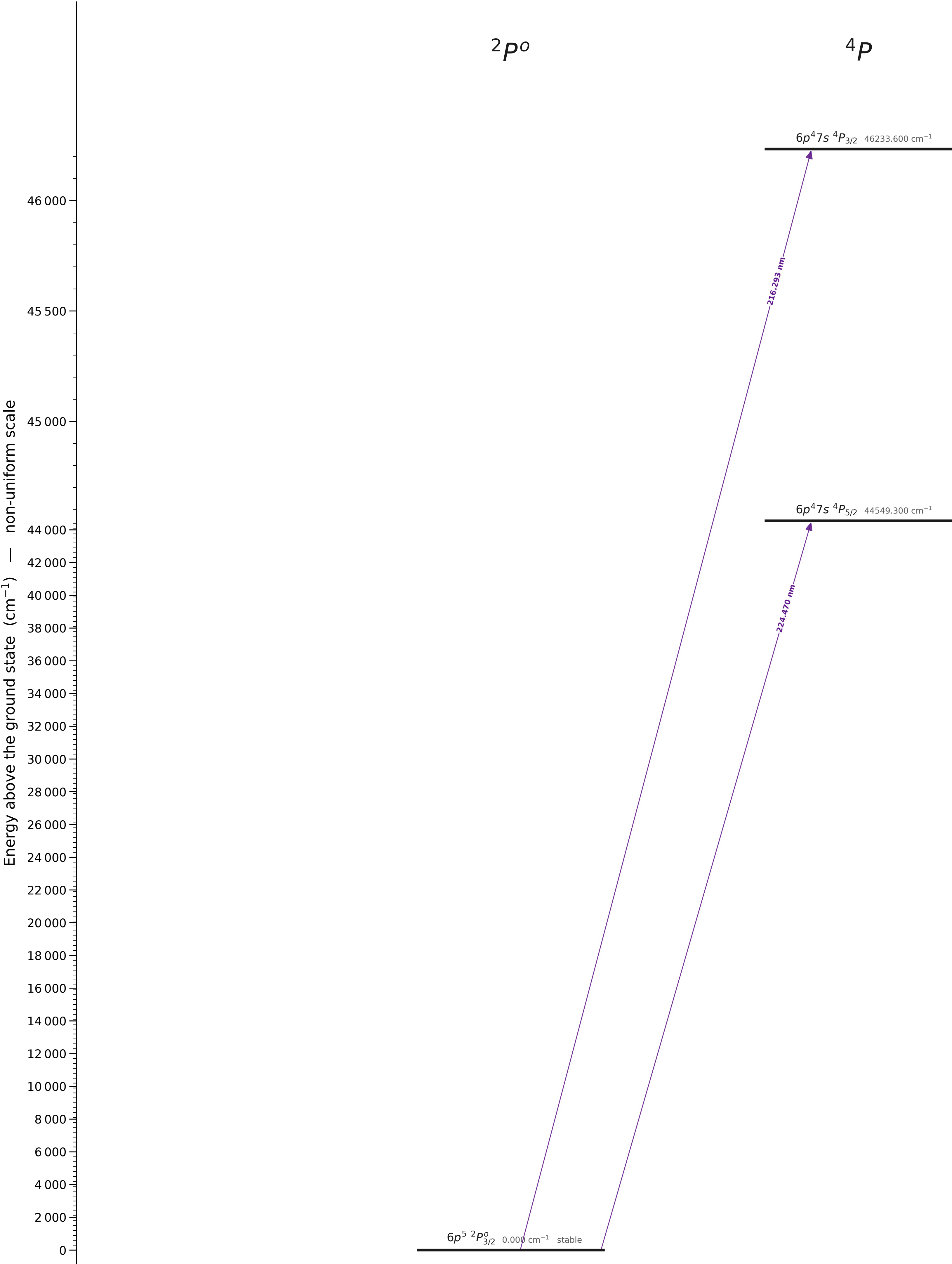


Astatine

At (At I). Strongest classified transitions below 2 μm (NIST): 3 levels, 2 lines.

At⁺ ionisation limit 75150.80 cm^{−1} = 9.31751 eV (133.066 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J e $\mathbf{r}$ J'⟩ (ea $\circ$)	A (s ^{−1})	Γ/2π up. (MHz)	branch (%)	Use
6p ⁵ 2P _{3/2} ^o – 6p ⁴ 7s 4P _{3/2}	216.2929	216.2250	1386.0485	–	–	–	–	first step of the ISOLDE-RILIS / TRIUMF-TRILIS three-colo...
6p ⁵ 2P _{3/2} ^o – 6p ⁴ 7s 4P _{5/2}	224.4704	224.4008	1335.5544	–	–	–	–	alternative first step for resonance ionization

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²⁰⁵At hyperfine structure (*I* = 9/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)				
6p ⁵ 2P _{3/2} ^o	676(100)	1835(775)	F=3: -4735.96	F=4: -3255.29	F=5: -257.58	F=6: +5021.75	
6p ⁴ 7s 4P _{3/2}	-365(3)	-367(50)	F=3: +2843.04	F=4: +1627.71	F=5: -120.83	F=6: -2555.50	

Measured A, B constants; sources in the data file.

Arrow label: vacuum wavelength | reduced dipole moment
Matrix elements are measured or NIST values; * =
Arrow width grows with the Einstein coefficient A.
Level caption: energy (NIST, or a measurement of th

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
6p ⁵ 2P _{3/2} ^o – 6p ⁴ 7s 4P _{3/2}	216.293	197-205	-7190(750)
6p ⁵ 2P _{3/2} ^o – 6p ⁴ 7s 4P _{3/2}	216.293	197m-205	-2975(750)
6p ⁵ 2P _{3/2} ^o – 6p ⁴ 7s 4P _{3/2}	216.293	198-205	-8731(750)
6p ⁵ 2P _{3/2} ^o – 6p ⁴ 7s 4P _{3/2}	216.293	203-205	-3437(900)
6p ⁵ 2P _{3/2} ^o – 6p ⁴ 7s 4P _{3/2}	216.293	207-205	2201(900)
6p ⁵ 2P _{3/2} ^o – 6p ⁴ 7s 4P _{3/2}	216.293	209-205	6232(750)
6p ⁵ 2P _{3/2} ^o – 6p ⁴ 7s 4P _{3/2}	216.293	211-205	9423(900)

Shift = ν(first isotope) – ν(second isotope).